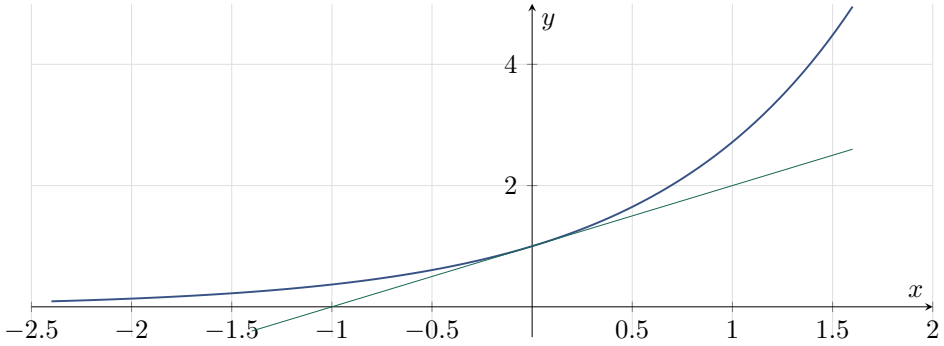
9.1 Defining a new function

Definition and results

The exponential function is the unique function f differentiable on $\mathbb{R}$ satisfying $f' = f$ and $f(0) = 1$. Its existence and uniqueness are assumed. It is denoted $\exp$, then e^x with $e = \exp(1) \approx 2{,}71828$. The number e is a constant; x is the variable. The exponential is strictly positive.

Status and justification. Existence and uniqueness accepted, as in the syllabus. The positivity is deduced, after the algebraic properties assumed below, from $\exp(x) = \exp(\frac{x}{2})^2$ and $\exp(x)\exp(-x) = 1$.

Worked example. $\exp(0) = 1$, $\exp(1) = e$, $\exp'(0) = 1$. The tangent at 0 is therefore $y = 1 + x$. The notation e^x extends the properties of powers to any real exponent.



e^x (blue) and its tangent $1 + x$ at 0 (green).

Watch out

$e^0 = 1$, not 0. The function never takes the value 0, even for negative x.

Immediate application. What is the equation of the tangent to e^x at 1?

9.2 Algebraic properties and comparisons

Definition and results

For all real numbers x, y, $e^{x+y} = e^x e^y$, $e^{-x} = \frac{1}{e^x}$, $e^{x-y} = \frac{e^x}{e^y}$. For integer n, $(e^x)^n = e^{nx}$. The function is strictly increasing on $\mathbb{R}$, therefore $e^u = e^v \iff u = v$ and $e^u < e^v \iff u < v$.

Status and justification. The addition formula is admitted in the common core; it can be explored further via uniqueness. Positivity and $\exp' = \exp > 0$ give strict monotonicity. This justifies equivalences for equations and inequalities.

Worked example. $e^{2x}e^{1-x} = e^{x+1}$. $e^{3x-2} = e^{x+4}$ is equivalent to 3 $x - 2 = x + 4$, so x=3. $e^{2x} \leq 1 = e^0$ is equivalent to $x \leq 0$.

Watch out

e^{x+y} is a product, never in general $e^x + e^y$. We do not solve $e^x = 2$ with a logarithm not studied in Première.

Immediate application. Simplify $\frac{e^{3x}}{e^{x-1}}$; solve $e^{x+2} > e^{2x}$.

9.3 Differentiate and study

Definition and results

For real a, $t \mapsto e^{at}$ is differentiable and its derivative is ae^{at}. For $e^{at+b} = e^b e^{at}$ the derivative is ae^{at+b}. The product and quotient rules apply with derivatives already known. The sign of the exponential often simplifies the study of the sign of the derivative.

Status and justification. The differentiation formula for e^{at} is admitted; the case with b follows from it by taking out the constant factor e^b. The general chain rule is not needed here.

Worked example. For $f(x) = (x+1)e^{-x}$, $f'(x) = e^{-x} - (x+1)e^{-x} = -xe^{-x}$. Since $e^{-x} > 0$, the derivative has the sign of $-x$: global maximum 1 at 0.

Watch out

Do not forget the factor a or the minus sign in e^{-x}. The exponential factor is positive, not necessarily increasing.

Immediate application. Differentiate $g(t) = 3e^{-2t}$ and study how it varies.

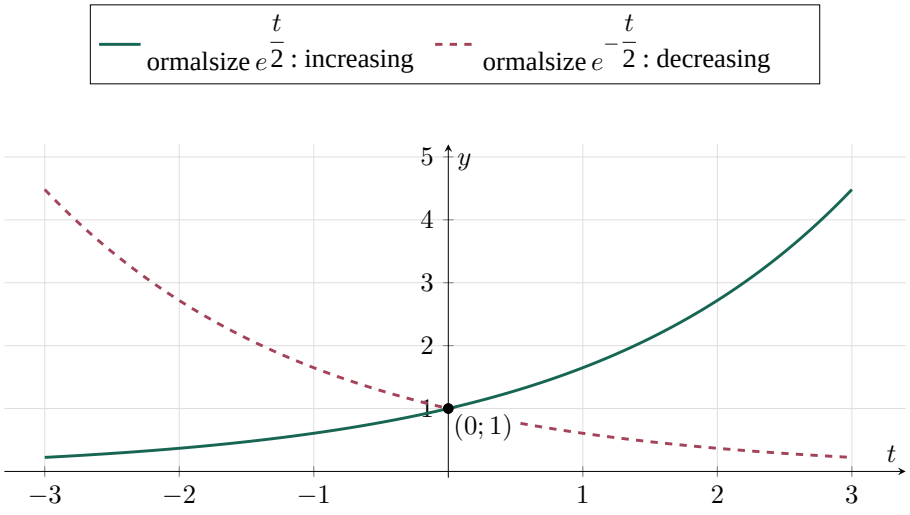
9.4 From discrete to continuous

Definition and results

A model $Q(t) = Q_0 e^{kt}$, with $Q_0 > 0$, represents growth if $k > 0$, a decrease if $k < 0$, a constant if $k = 0$. For a fixed step h, $\dfrac{Q(t+h)}{Q(t)} = e^{kh}$. The values at times integers form a geometric sequence with common ratio e^k. The unit of k is the reciprocal of the unit of time.

Status and justification. The ratio comes from $e^{k(t+h)} = e^{kt}e^{kh}$. The variation is deduced from $Q\prime(t) = kQ_0e^{kt}$ and the positivity of the remaining factor.

Worked example. $Q(t) = 100e^{-0{,}2t}$, t in hours: $Q(0) = 100$, $\dfrac{Q(t+1)}{Q(t)} = e^{-0{,}2}$. A simulation can look for the first whole number of hours at which Q falls below 50; the exact equation is not solved by logarithm in this chapter.



The curves corresponding to $k = \dfrac{1}{2}$ and $k = -\dfrac{1}{2}$ are symmetric about the vertical axis. The line styles also distinguish them without colour.

Watch out

The relative rate per unit of discrete time is $e^k - 1$, not exactly k; they are close only for small k.

Immediate application. For $Q(t) = 5e^{0{,}3t}$, express $Q(n)$ and $\dfrac{Q(t+2)}{Q(t)}$.

9.5 Build numerically and prove an inequality

Definition and results

To approximate the solution of $y' = y,\ y(0) = 1$ with one step h, Euler replaces the increment by $hy : y_{k+1} = (1+h)y_k$. On [0; 1] with $h = \frac{1}{n}$, the approximation of e is $\left(1+\frac{1}{n}\right)^n$. This construction is experimental. The inequality $e^x \geq 1 + x$ is obtained by studying a function.

Status and justification. For $g(x) = e^x - 1 - x,\ g'(x) = e^x - 1$, negative if x<0 and positive if x>0. The minimum is $g(0) = 0$, which proves the inequality; equality only at 0. Convergence of Euler is not proved here.

Worked example. With n=2, Euler's method gives $y_0 = 1, y_1 = 1{,}5, y_2 = 2{,}25$. With n=10, $(1\ ,1)^{10} \approx 2{,}59374$. We compare with e; we do not confuse an approximate value with the definition of the function.

Watch out

A graph showing a curve above a line does not prove the inequality for all the real numbers.

Immediate application. Show that $e^{2x} \geq 1 + 2x$ for all real x.